

Vaghesan Sundaram

(917) 297-6785 | vaghesan2@gmail.com | GitHub | LinkedIn | Portfolio

Education

University of Maryland, College Park

B.S. Computer Science, Cybersecurity Focus; B.S. Mathematics; ACES Cybersecurity Minor

Expected Fall 2027

College Park, MD

Relevant Coursework: Data Structures, Algorithms, Computer Systems, Statistics, Cryptography,

Network Security, HACS408T Penetration Testing

Middlesex College

A.S. Mathematics

June 2024

Edison, NJ

Certification & Technical Skills

CompTIA Security+ (SY0-701), issued March 2026; expires March 2029.

Languages: Python, Java, C++, C, Assembly, Rust

Security: Secure code review, vulnerability analysis, authentication and authorization review, evidence and severity triage, remediation reporting

Engineering: Pydantic, pytest, Hypothesis, mypy, GitHub Actions, Flask, SQLite, Git, Linux

Tools: Semgrep, Wireshark, PowerShell; Burp Suite and Nmap in controlled labs

Experience

Cybersecurity Intern

Jun 2026 to Present

Redmind Technologies

Remote

- Assessed one public-facing client site and documented 7 findings spanning data exposure, authentication flows, configuration, and dependency risk.
- Combined Semgrep and manual review with evidence capture, severity triage, and remediation mapping to produce developer-facing guidance.
- Presented the findings in a written report and slide deck to developers, explaining risk, evidence, and recommended fixes.

Information Technology Intern

Jun to Aug 2025

New Jersey Courts

Toms River, NJ

- Supported production IT systems across court departments through troubleshooting, access workflows, user support, documentation, and escalation.
- Documented recurring issues and support procedures for repeatable troubleshooting and clearer team handoffs.

Selected Projects

Agent Least Privilege

GitHub

- Built a Python/Pydantic default-deny authorization monitor to bind tool calls to task-scoped functions, resources, arguments, approvals, and call limits, with append-only audit events and deterministic replay.
- Validated it with 93 tests, Hypothesis property checks, strict mypy, and Windows/Linux CI; in 60 episodes, task-conditioned enforcement produced 0/10 unauthorized state-changing effects versus 4/10 unrestricted and 6/10 with an allowlist.

CTI Claim Provenance

GitHub

- Designed a 64-question point-in-time CTI benchmark across 24 dependency groups with cutoff and authority rules, source hashes, typed expected answers, abstention cases, human review, and split isolation.
- Published a privacy-safe 192-cell comparison; constraints raised evidence binding from 83.3% to 86.5% but reduced exact-answer matching from 33.3% to 26.0%, separating evidence selection from answer construction.

Keystroke Authentication System

GitHub

- Built a two-service Flask authentication system with SQLite persistence, typing-cadence validation, JWT-bound flows, bcrypt fallback, rate limiting, and single-use server-to-server code exchange.
- Added an end-to-end smoke test covering account creation, three-sample enrollment, and authentication.